

## Project Thesis 1

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### Observability for AI Agents: Systematic Analysis and Evaluation of Open-Source Tooling

#### Introduction

AI agents increasingly orchestrate tools, maintain memory, and execute long-running or autonomous workflows. As a result, their internal behavior and system-level impact become difficult to understand, debug, and govern without robust observability mechanisms. Emerging standards—such as OpenTelemetry GenAI semantic conventions and the OWASP Agent Observability Standard—alongside a rapidly growing ecosystem of open-source tooling, offer new possibilities but also introduce fragmentation and uncertainty for engineering teams. The stochastic nature of LLM outputs means that traditional "pass/fail" monitoring is insufficient; we require state-aware observability that captures the reasoning trace and the evolution of the agent's internal context window.

This project focuses explicitly on observability for AI agents: identifying what should be measured, how agent-based systems can be instrumented, how existing open-source solutions compare, and where current approaches fall short—particularly in multi-agent and distributed settings.

#### Tasks

- Observability requirements for AI agents:** Define observability requirements specific to agent-based systems, including:
  - Tracing of agent reasoning steps, chain-of-thought (CoT) trajectories, and tool usage
  - Agent-to-agent communication, message passing protocols, and shared blackboard state coordination
  - Quality, performance, and safety metrics (e.g., token usage/cost attribution, hallucination rates, and latency per reasoning step)
  - Alignment, governance, and compliance-related signals
- Analysis of existing open-source solutions:** Investigate current open-source observability tooling by:

- a. Identifying relevant frameworks, libraries, and platforms (differentiating between OTel-native tools like OpenLLMetry and evaluation-heavy tools like Langfuse)
  - b. Analyzing available integration points with agent frameworks (e.g., LangChain) and runtimes
  - c. Selecting a subset of tools for detailed comparison and implementation
  - d. Conducting an in-depth comparison across dimensions such as maturity, standards alignment, usability, and coverage of agent-specific signals
3. **Implementation:**
- a. Design and implement a minimal, framework-agnostic agent workload (e.g., a CLI or HTTP-based agent performing tool-augmented tasks) as:
    - i. A single agent
    - ii. A small multi-agent system using simple in-process message routing
  - 4. Possibly in a simple distributed setting over multiple nodes
  - 5. Instrument the workload using OpenTelemetry and one or two selected observability tools to produce traces, metrics, and logs, with particular emphasis on:
    - i. Correlation across agents, tools, and external services
    - ii. End-to-end visibility of agent workflows
6. **Patterns, best practices, limitations, and outlook:**
- a. Derive architectural patterns and best practices for observability in AI agent systems
7. Identify and document key limitations of current approaches, including:
- i. Inconsistent or incomplete semantic conventions
  - ii. Lack of standardized agent identifiers
  - iii. Trace and metric explosion in multi-agent topologies
  - iv. Missing or immature quality and safety metrics (e.g., lack of real-time detection for recursive agent loops)
  - v. Challenges related to data redaction, privacy, and compliance
- b. Propose practical guidelines for production-grade agent observability aligned with emerging standards
  - c. Outline future research directions, such as:
    - i. Standardized agent-level semantic conventions
    - ii. Observability-driven evaluation and feedback loops
    - iii. Observability for agent meshes and A2A/MCP-based ecosystems

## Evaluation

- Written report, incl. management review
- Written summary of the analysis and results in a publishable format
- GitHub Implementation of the project
- Presentation (15 minutes)

## Literature

- [1] S. Murugesan, "The Rise of Agentic AI: Implications, Concerns, and the Path Forward," in IEEE Intelligent Systems, vol. 40, no. 2, pp. 8-14, March-April 2025, doi: 10.1109/MIS.2025.3544940.
- [2] D. Moshkovich, and Sergey Zeltyn. *"Taming uncertainty via automation: Observing, analyzing, and optimizing agentic ai systems."* *arXiv preprint arXiv:2507.11277* (2025).
- [3] OpenTelemetry Community, "Semantic Conventions for Generative AI Operations," GitHub/OTel Docs.
- [4] Anthropic, "Model Context Protocol (MCP) Specification," modelcontextprotocol.io.
- [5] A. Shaik, *"AI's Next Leap: Agentic Intelligence"* Communications of ACM.
- [6] S. Falconer. *Agentic Mesh — The Future of Enterprise Agent Ecosystems*, Medium, June 2025.
- [7] AGNTCY Project, <https://github.com/agntcy>